

Uncertain knowledge and Reasoning

Exercise (Joint Distribution)

- At a certain university, 4% of men are over 6 feet tall and 1% of women are over 6 feet tall. The total student population is divided in the ratio 3:2 in favour of women. If a student is selected at random from among all those over six feet tall, what is the probability that the student is a woman?

Let $M = \{\text{Student is Male}\}$, $F = \{\text{Student is Female}\}$, (note that M and F partition the sample space of students), $T = \{\text{Student is over 6 feet tall}\}$. We know that $P(M) = 2/5$, $P(F) = 3/5$, $P(T|M) = 4/100$ and $P(T|F) = 1/100$. We require $P(F|T)$. Using Bayes' Theorem we have:

$$\begin{aligned} P(F|T) &= \frac{P(T|F)P(F)}{P(T|F)P(F) + P(T|M)P(M)} \\ &= \frac{\frac{1}{100} \times \frac{3}{5}}{\frac{1}{100} \times \frac{3}{5} + \frac{4}{100} \times \frac{2}{5}} \\ &= \frac{3}{11} \end{aligned}$$

The Joint Probability Distribution

- Once you have the joint probability distribution, you can calculate any probability involving A , B , and C

A	B	C	P(A,B,C)
false	false	false	0.1
false	false	true	0.2
false	true	false	0.05
false	true	true	0.05
true	false	false	0.3
true	false	true	0.1
true	true	false	0.05
true	true	true	0.15

Examples of things you can compute:

- $P(A=true)$ = sum of $P(A,B,C)$ in rows with $A=true$
- $P(A=true, B = true \mid C=true) =$

$$P(A = true, B = true, C = true) / P(C = true)$$

The Problem with the Joint Distribution

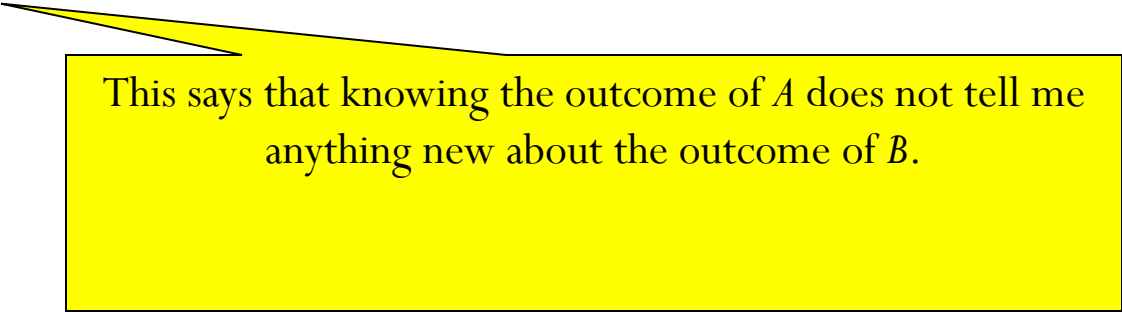
- Lots of entries in the table to fill up!
- For k Boolean random variables, you need a table of size 2^k
- How do we use fewer numbers? Need the concept of independence

A	B	C	P(A,B,C)
false	false	false	0.1
false	false	true	0.2
false	true	false	0.05
false	true	true	0.05
true	false	false	0.3
true	false	true	0.1
true	true	false	0.05
true	true	true	0.15

Independence

Variables A and B are independent if any of the following hold:

- $P(A,B) = P(A) P(B)$
- $P(A \mid B) = P(A)$
- $P(B \mid A) = P(B)$



This says that knowing the outcome of A does not tell me anything new about the outcome of B .

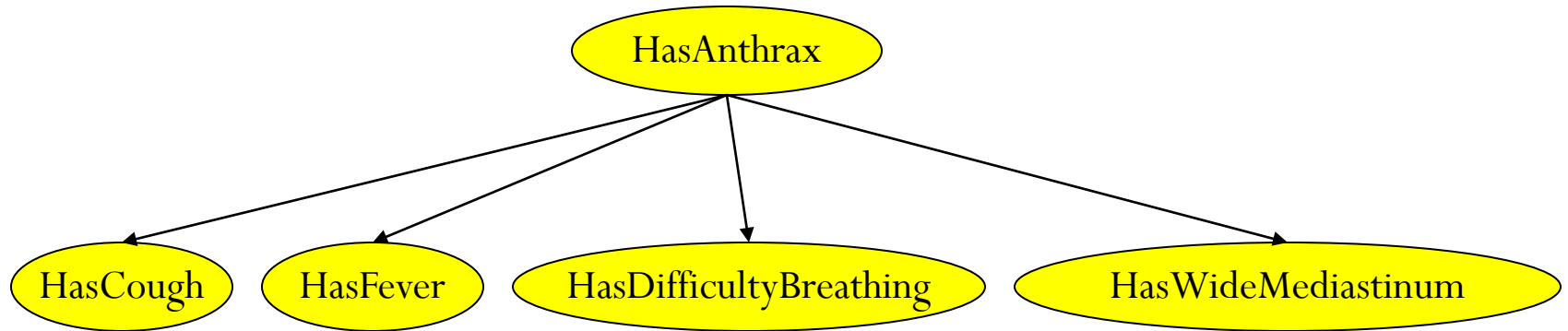
Conditional Independence

Variables A and B are conditionally independent given C if any of the following hold:

- $P(A, B \mid C) = P(A \mid C) P(B \mid C)$
- $P(A \mid B, C) = P(A \mid C)$
- $P(B \mid A, C) = P(B \mid C)$

Knowing C tells me everything about B . I don't gain anything by knowing A (either because A doesn't influence B or because knowing C provides all the information knowing A would give)

Bayesian Networks

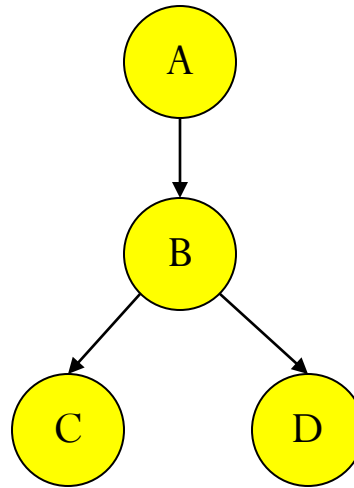


- In the opinion of many AI researchers, Bayesian networks are the most significant contribution in AI in the last 10 years
- They are used in many applications eg. spam filtering, speech recognition, robotics, diagnostic systems and even syndromic surveillance

A Bayesian Network

A Bayesian network is made up of:

1. A Directed Acyclic Graph



2. A set of tables for each node in the graph

A	P(A)
false	0.6
true	0.4

A	B	P(B A)
false	false	0.01
false	true	0.99
true	false	0.7
true	true	0.3

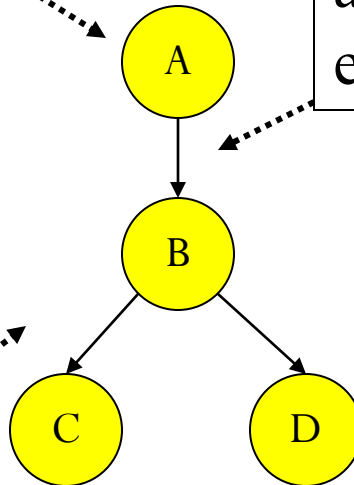
B	D	P(D B)
false	false	0.02
false	true	0.98
true	false	0.05
true	true	0.95

B	C	P(C B)
false	false	0.4
false	true	0.6
true	false	0.9
true	true	0.1

A Directed Acyclic Graph

Each node in the graph is a random variable

A node X is a parent of another node Y if there is an arrow from node X to node Y
eg. A is a parent of B



Informally, an arrow from node X to node Y means X has a direct influence on Y

A Set of Tables for Each Node

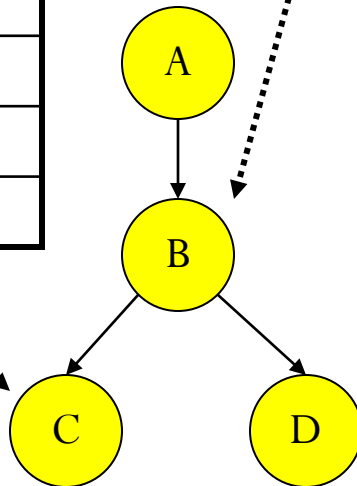
A	P(A)
false	0.6
true	0.4

A	B	P(B A)
false	false	0.01
false	true	0.99
true	false	0.7
true	true	0.3

B	C	P(C B)
false	false	0.4
false	true	0.6
true	false	0.9
true	true	0.1

Each node X_i has a conditional probability distribution $P(X_i \mid \text{Parents}(X_i))$ that quantifies the effect of the parents on the node

The parameters are the probabilities in these conditional probability tables (CPTs)



B	D	P(D B)
false	false	0.02
false	true	0.98
true	false	0.05
true	true	0.95

A Set of Tables for Each Node

Conditional Probability
Distribution for C given B

B	C	P(C B)
false	false	0.4
false	true	0.6
true	false	0.9
true	true	0.1

For a given combination of values of the parents (B in this example), the entries for $P(C=\text{true} \mid B)$ and $P(C=\text{false} \mid B)$ must add up to 1
eg. $P(C=\text{true} \mid B=\text{false}) + P(C=\text{false} \mid B=\text{false}) = 1$

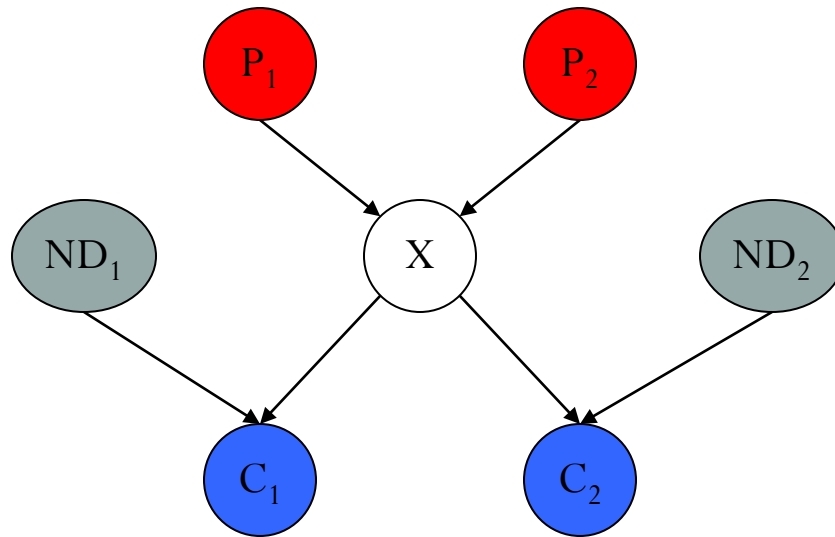
Bayesian Networks

Two important properties:

1. Encodes the conditional independence relationships between the variables in the graph structure
2. Is a compact representation of the joint probability distribution over the variables

Conditional Independence

The Markov condition: given its parents (P_1, P_2), a node (X) is conditionally independent of its non-descendants (ND_1, ND_2)



The Joint Probability Distribution

Due to the Markov condition, we can compute the joint probability distribution over all the variables $X_1, \dots, X_n$ in the Bayesian net using the formula:

$$P(X_1 = x_1, \dots, X_n = x_n) = \prod_{i=1}^n P(X_i = x_i \mid \text{Parents}(X_i))$$

Where $\text{Parents}(X_i)$ means the values of the Parents of the node X_i with respect to the graph

Using a Bayesian Network Example

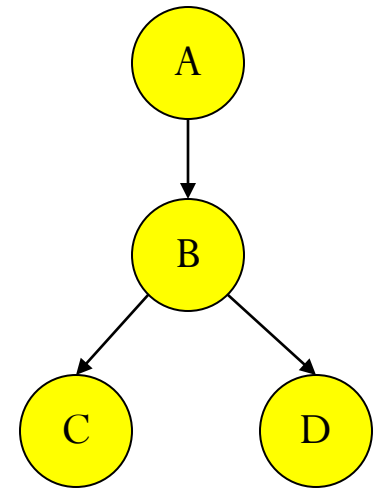
Using the network in the example, suppose you want to calculate:

$$P(A = \text{true}, B = \text{true}, C = \text{true}, D = \text{true})$$

$$= P(A = \text{true}) * P(B = \text{true} \mid A = \text{true}) *$$

$$P(C = \text{true} \mid B = \text{true}) P(D = \text{true} \mid B = \text{true})$$

$$= (0.4) * (0.3) * (0.1) * (0.95)$$



Using a Bayesian Network Example

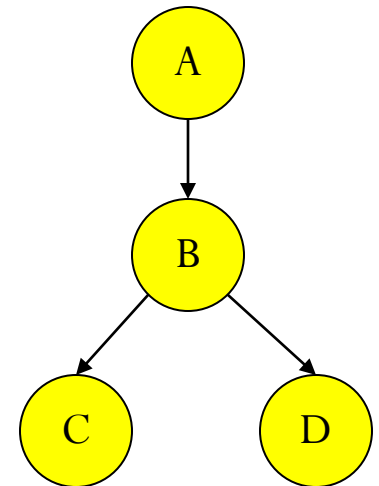
Using the network in the example, suppose you want to calculate:

$$\begin{aligned} &P(A = \text{true}, B = \text{true}, C = \text{true}, D = \text{true}) \\ &= P(A = \text{true}) * P(B = \text{true} \mid A = \text{true}) * \\ &\quad P(C = \text{true} \mid B = \text{true}) P(D = \text{true} \mid B = \text{true}) \\ &= (0.4) * (0.3) * (0.1) * (0.95) \end{aligned}$$

This is from the
graph structure



These numbers are from the
conditional probability tables



Example: Traffic II

- Let's build a causal graphical model!

- Variables

- T: Traffic
- R: It rains
- L: Low pressure
- D: Roof drips
- B: Ballgame
- C: Cavity



Example: Traffic II

- Let's build a causal graphical model!

- Variables

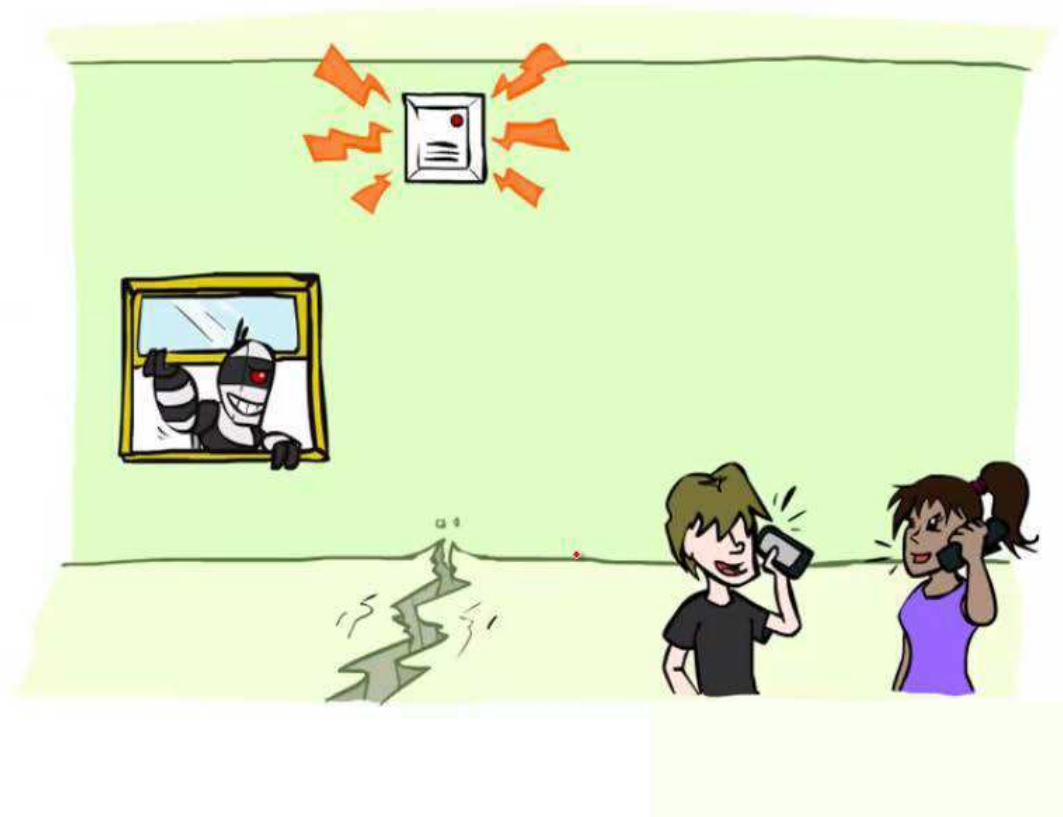
- T: Traffic
- R: It rains
- L: Low pressure
- D: Roof drips
- B: Ballgame
- C: Cavity



Example: Burglar Alarm

- I have a burglar alarm that is sometimes set off by minor earthquakes. My two neighbors, John and Mary, promised to call me at work if they hear the alarm
 - Example inference task: suppose Mary calls and John doesn't call. What is the probability of a burglary?
- What are the random variables?
 - *Burglary, Earthquake, Alarm, JohnCalls, MaryCalls*
- What are the direct influence relationships?
 - A burglar can set the alarm off
 - An earthquake can set the alarm off
 - The alarm can cause Mary to call
 - The alarm can cause John to call

Example: Alarm Network



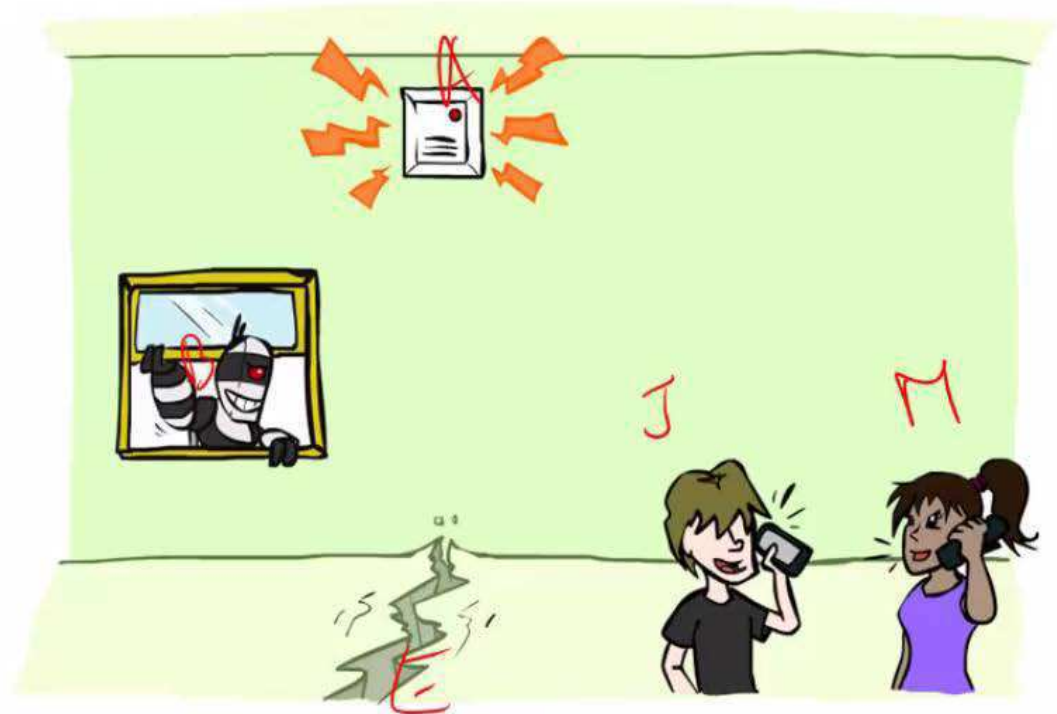
Example: Alarm Network



Example: Alarm Network

■ Variables

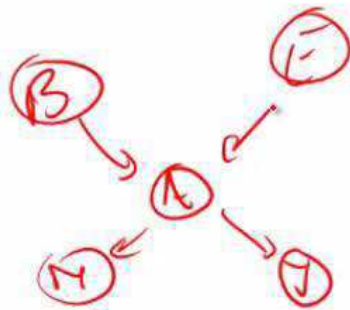
- B: Burglary
- A: Alarm goes off
- M: Mary calls
- J: John calls
- E: Earthquake!



Example: Alarm Network

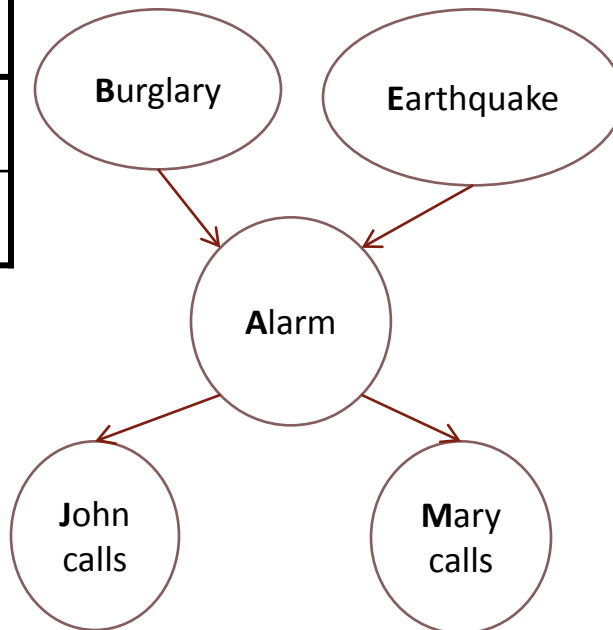
■ Variables

- B: Burglary
- A: Alarm goes off
- M: Mary calls
- J: John calls
- E: Earthquake!



Example: Alarm Network

B	P(B)
+b	0.001
-b	0.999



E	P(E)
+e	0.002
-e	0.998



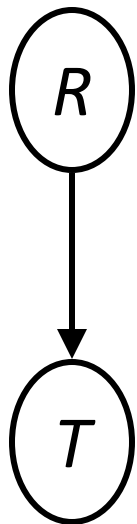
A	J	P(J A)
+a	+j	0.9
+a	-j	0.1
-a	+j	0.05
-a	-j	0.95

A	M	P(M A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99

B	E	A	P(A B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

Example: Traffic

- Causal direction



$P(R)$

+r	1/4
-r	3/4

$P(T|R)$

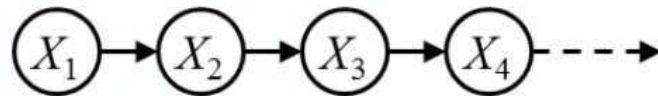
+r	+t	3/4
	-t	1/4

-r	+t	1/2
	-t	1/2



Markov Models

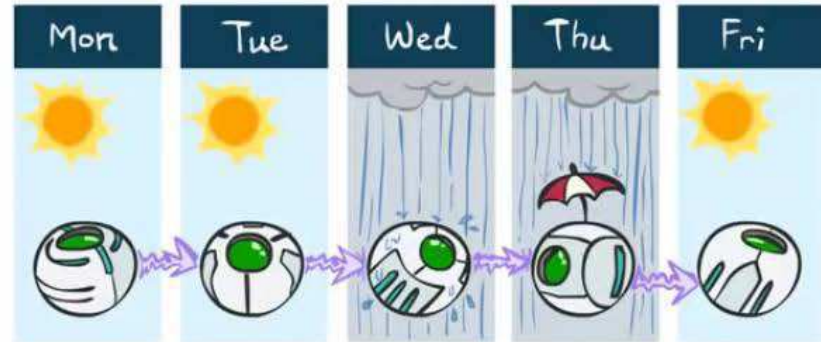
- Value of X at a given time is called the **state**



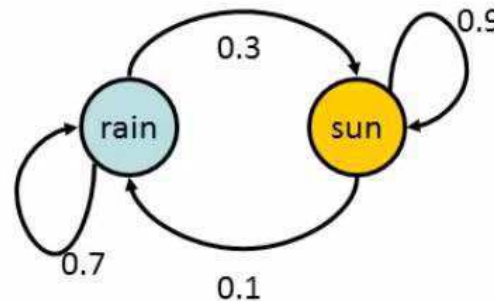
Example Markov Chain: Weather

- States: $X = \{\text{rain}, \text{sun}\}$
- Initial distribution: 1.0 sun
- CPT $P(X_t | X_{t-1})$: 

X_{t-1}	X_t	$P(X_t X_{t-1})$
sun	sun	0.9
sun	rain	0.1
rain	sun	0.3
rain	rain	0.7



Two new ways of representing the same CPT



Homework

- Exercises :

14.3, 14.11, 15.2 (a) and 15.3 (a)

from textbook [Peter Norvig and Russel]